Evaluating algorithms for antisemitism detection

Kornstein Lab for Antisemitism Research

Research proposal

Abstract

This proposed study evaluates how algorithms distinguish antisemitic expression from quotation, criticism, and other contextual uses. It examines classification errors, consistency, and performance across different kinds of text.

1 Introduction

We propose a contextual evaluation of algorithms for detecting antisemitism. The aim is to compare model outputs with reviewed annotations and examine failure modes across linguistic settings.

2 Evaluation framework

An illustrative classification measure is

$$F_1 = \frac{2PR}{P+R}, \qquad P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}.$$

Precision P and recall R summarize different error costs. We would report both with uncertainty and contextual error analysis, rather than rely on a single score. Undefined ratios require explicit handling.

3 Proposed methodology

1. Contextual annotation.
2. Algorithm benchmarking.
3. Error and robustness analysis.

4 Limitations

Annotation judgments and corpus selection may limit transfer across languages and contexts. Model performance remains to be measured; this proposal reports no benchmark results.